



VIRTUTOUR PLATFORM OVERVIEW

VirtuTour Pipeline Description and Feature Scope

A hotel room preview layer - from room inventory to live 360 degree VR booking confidence.

VirtuTour is an immersive hotel room and venue preview platform for hospitality teams. It lets guests inspect suites, lobbies, pools, ballrooms, and room categories in a browser or VR headset before they commit to a booking. The platform combines 360 degree room capture, Unreal Engine 5 scene production, pixel streaming, booking-session orchestration, headset sync, hotel inventory mapping, and analytics into one connected pre-booking workflow.

Hotel VR preview

360 degree rooms

UE5 pixel streaming

Virtual bookings

Headset sync

8K fidelity stream

From room selection to booking confirmation, in one preview loop

For hotels, resorts, travel agencies, and premium hospitality booking teams.

- 1 Map hotel rooms, suites, public spaces, and booking categories to verified 360 degree or UE5-ready visual assets.
- 2 Launch a browser or headset preview so the guest can inspect the exact room class, view, layout, and amenity context.
- 3 Convert the preview into a confirmed reservation with booking status, session logs, and hotel-side stream monitoring.

DATE

2026-06-18

PRODUCT

VirtuTour

CATEGORY

Hotel VR room preview, virtual booking, and UE5 pixel-streaming hospitality platform

Platform architecture, guest workflow, hotel operations model, stream configuration, and full capability description - prepared for technical, commercial, and hospitality stakeholder review.

Platform overview

Six properties that turn room browsing into booking confidence

The booking journey becomes visual, inspectable, and trustable. Guests do not rely only on staged photos; they can walk the room, compare spaces, and confirm the stay with fewer surprises.

01

Pre-booking room certainty

Guests can inspect a suite or room category before paying, reducing mismatch between hotel photos, guest expectations, and the real room experience.

02

Browser and headset access

Preview sessions can run as a browser pixel stream or as a headset workflow for hotel sales desks, premium guests, and travel partners.

03

Hotel inventory awareness

Room names, suites, venues, bookings, availability states, and preview assets are connected so the preview matches the category being sold.

04

Live stream operations

Hotel operators can monitor active streams, bitrate, latency, viewer load, stream status, and signaling-server health from one dashboard.

05

Premium hospitality positioning

Luxury suites, banquet spaces, poolside experiences, and destination rooms become immersive products instead of static gallery images.

06

Conversion feedback loop

Every preview can be tied to a booking request, confirmation status, guest session, and hotel-side analytics for measurable sales impact.

— BUILT FOR

Hospitality teams that sell experience before arrival

VirtuTour is designed for hotels and travel sellers where room quality, view, ambience, and space layout strongly affect the booking decision.

01

Luxury hotels

Show premium suites, executive floors, presidential rooms, and branded amenities before the guest commits.

02

Resorts and destinations

Preview ocean views, pool areas, villas, family rooms, honeymoon suites, and seasonal destination experiences.

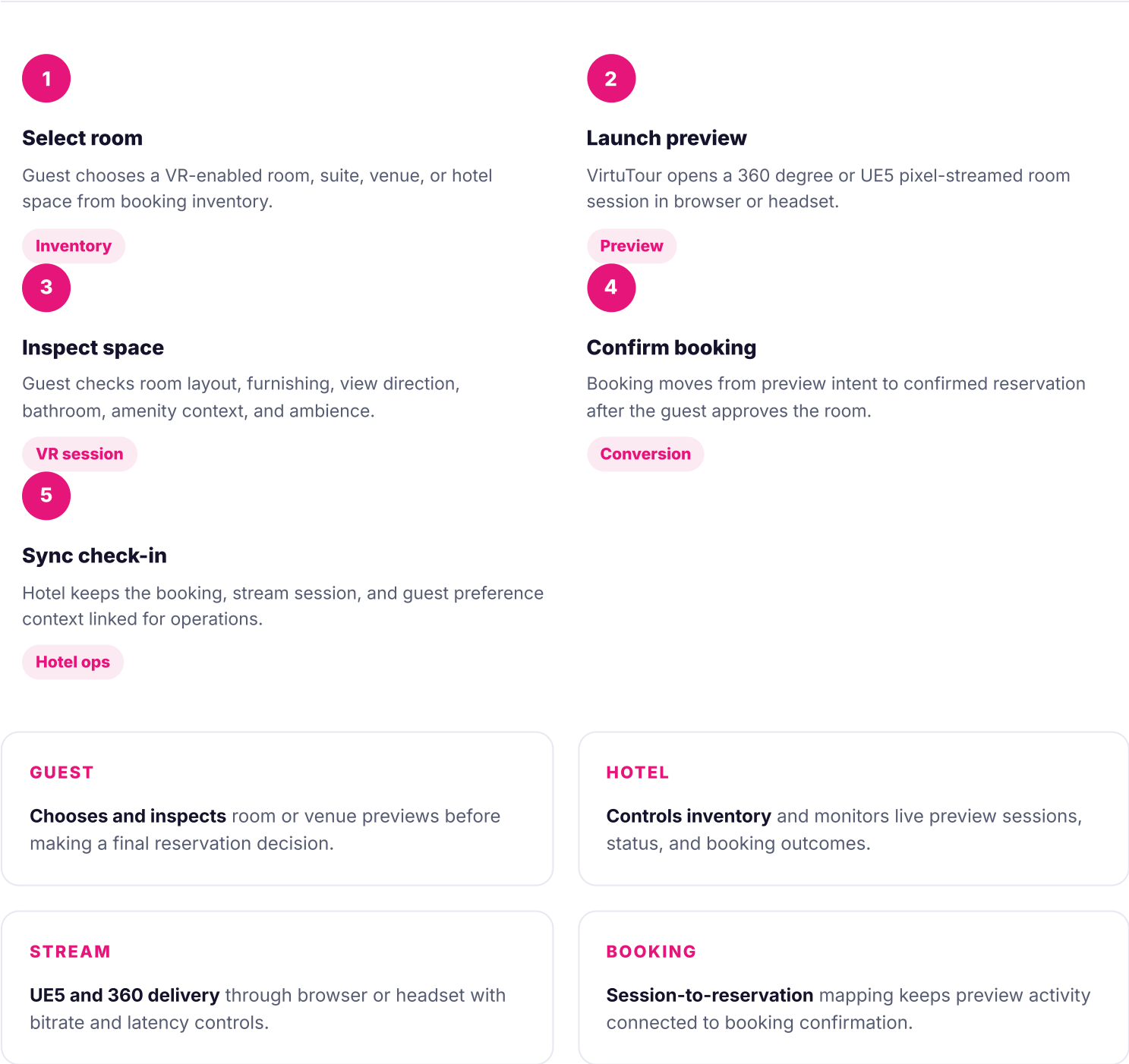
03

Event and banquet teams

Let clients inspect ballrooms, conference halls, wedding layouts, and lobby flow before shortlisting a venue.

The pre-booking path VirtuTour is built around

The core workflow starts before payment. The guest selects a room, launches a visual preview, inspects the experience, and confirms the booking after the room has been understood.



Six layers of the VirtuTour platform

VirtuTour behaves as a hospitality preview system, not only a media gallery. Every room asset, stream, booking, and guest session has a defined place in the pipeline.

1 Room and hotel inventory plane

ROOMS, SUITES, VENUES

- Room categories
- Hotel locations
- Availability states
- Preview eligibility

2 Capture and asset plane

360 AND MEDIA SOURCE

- 360 room footage
- Still images
- Suite thumbnails
- Venue captures

3 UE5 production plane

IMMERSIVE SCENE

- Lighting
- Materials
- Interactive room views
- 8K fidelity targets

4 Pixel streaming plane

BROWSER AND HEADSET

- WebRTC stream
- Signaling server
- Bitrate control
- Live status

5 Booking orchestration plane

PREVIEW TO RESERVATION

- Virtual bookings
- Pending states
- Confirmed states
- Guest session logs

6 Hotel operations plane

MONITORING AND ANALYTICS

- Viewer load
- Latency metrics
- Headset sync
- Conversion tracking

The stack VirtuTour is engineered around

The platform combines hotel media assets, interactive UE5 scenes, browser streaming, and booking workflow controls with an operations dashboard for hotel staff.

● Operational today ● In place ● Roadmap

Preview delivery

● Browser room preview

Guest can inspect the room without installing software

● VR headset workflow

Hotel-side or premium guest headset preview sessions

● 360 degree visual navigation

Room view, layout, ambience, and amenity inspection

● Multi-room comparison

Side-by-side comparison of room categories before booking

Streaming layer

● UE5 pixel streaming

Server-rendered hotel environments delivered to client devices

● Signaling server endpoint

Configurable WebRTC connection such as wss://ue5.virtutour.bd:8888

● Bitrate and codec controls

50 Mbps to 400 Mbps profiles, H.264/H.265/AV1 options

● Bangladesh hub routing

Dhaka primary hub, Chattogram and Sylhet edge-node model

Booking system

Virtual booking queue

Upcoming VR preview bookings visible to hotel staff

Confirmed and pending states

Reservation state connected to guest preview sessions

Session duration and guest metadata

Preview type, duration, guest identifier, and suite mapping

PMS integration

Hotel property-management-system sync as a production extension

Device operations

Headset sync

Hotel operator can sync headset sessions from the dashboard

Live metrics

Latency, throughput, vertex count, FPS, and active viewers

Stream status grid

Live, buffering, and idle stream state by hotel asset

Automated capacity alerts

Viewer load and stream quality alerts for hotel operations teams

Toolchain posture reflects the current VirtuTour dashboard and product direction. The user-facing goal is direct: before booking, the guest can check the room or hotel space in an immersive 360 degree or VR stream.

Contents

01	Product Summary	Capability
02	Product Positioning	Commercial
03	Platform Architecture - Six Layers	Technical
04	Guest Booking Workflow	Operational
05	Core Feature Set	Capability
06	Hotel Operations Model	Operations
07	Governance & Standards	Delivery
08	Final Product Description	Summary

Full capability description

The following pages describe VirtuTour as a hotel VR room preview and booking-confidence platform, organized for technical, commercial, and hospitality readers.

— PRODUCT SUMMARY

VirtuTour is a hotel interior explorer for VR and browser-based room previews. It helps guests examine a room, suite, lobby, poolside area, ballroom, or other bookable hospitality space before confirming a reservation.

The core promise is simple: before booking, the user can check the hotel room in a 360 degree or VR view. That preview can run as an 8K fidelity pixel stream, a headset session, or a browser-based visual tour depending on the room asset and hotel workflow.

The product is built around hotel operations, not only guest browsing. Staff can manage previewable spaces, monitor active streams, review upcoming virtual bookings, sync headset devices, and connect preview sessions to confirmed or pending reservations.

— PRODUCT POSITIONING

VirtuTour sits between hotel marketing, room booking, and immersive media delivery. Traditional hotel websites rely on galleries that often hide room scale, view quality, bathroom layout, and ambience. VirtuTour makes those details inspectable before payment.

Market position

For premium hospitality, the room is the product. VirtuTour gives hotels a way to sell that product with proof: the guest can experience the room class before arrival, while the hotel can track which preview sessions convert into bookings.

— PLATFORM ARCHITECTURE - SIX LAYERS

1. Hotel Inventory Layer

- Hotel, room, suite, venue, and category records
- Preview eligibility by room type
- Availability and booking status awareness
- Hotel operator ownership

2. Capture and Asset Layer

- 360 degree room capture
- Still images and preview thumbnails
- Room category media archive
- Venue and amenity visual packages

3. Immersive Production Layer

- UE5 room or venue scene production
- Lighting, material, and ambience tuning
- Interactive room-view mechanics
- 8K fidelity targets for premium spaces

4. Streaming Layer

- WebRTC pixel streaming
- Signaling server configuration
- Bitrate, codec, and region settings
- Browser and headset playback

5. Booking Layer

- Virtual room booking queue
- Preview session tied to reservation intent
- Confirmed and pending states
- Guest, duration, room, and preview metadata

6. Operations Layer

- Stream health dashboard
- Latency, throughput, FPS, and viewer metrics
- Headset sync
- Conversion and usage analytics roadmap

GUEST BOOKING WORKFLOW

STEP	ACTIVITY	OWNER	OUTCOME
01	Select suite, room category, or hotel space from VR-enabled inventory	Guest / booking agent	Preview candidate selected
02	Launch browser or headset preview session	VirtuTour stream layer	360 degree or UE5 preview active
03	Inspect room layout, view, furnishing, bathroom, amenity context, and ambience	Guest	Room understood before payment
04	Confirm or reject the booking after immersive preview	Guest / booking system	Reservation status updated
05	Sync booking, session record, and check-in context	Hotel operator	Hotel has a traceable preview-to-booking record

CORE FEATURE SET

Hotel previews

- Room and suite previews
- Lobby, poolside, and ballroom previews
- Live preview cards

- Search by hotel, room, suite, or stream

VR and 360 viewing

- 360 degree room inspection
- VR headset launch
- Browser preview mode
- Stream modal with live HUD

Pixel streams

- UE5 pixel streaming
- 8K fidelity stream labels
- FPS, bitrate, latency, throughput
- Live, buffering, idle status

Bookings

- Upcoming virtual bookings
- Full booking list
- Confirmed and pending states
- Preview session to booking confirmation

Stream configuration

- Signaling URL management
- Target bitrate selection
- Codec selection
- Dhaka primary hub and edge region model

Hotel operations

- Operator dashboard
- Headset sync
- Notifications and support surface
- Preview-to-conversion analytics roadmap

— HOTEL OPERATIONS MODEL

OPERATING AREA	VIRTUTOUR BEHAVIOR	BUSINESS VALUE
Room sales	Guests inspect a realistic visual experience before committing	Higher confidence and fewer expectation gaps
Premium upsell	Suite, view, and amenity differences become visible	Easier upgrade from standard room to premium category
Event sales	Ballrooms and venue layouts can be previewed remotely	Faster shortlisting for events and corporate clients
Travel agents	Agents can launch guided previews for remote customers	More persuasive room recommendation workflow

OPERATING AREA	VIRTU TOUR BEHAVIOR	BUSINESS VALUE
Hotel monitoring	Staff see stream state, viewer load, and booking queue	Operational control over guest preview sessions

GOVERNANCE & STANDARDS

RULE	WHY IT EXISTS	ENFORCEMENT
Preview must match room category	Guest trust depends on visual accuracy	Required
Booking states stay visible	Staff must separate confirmed and pending reservations	Required
Stream health is monitored	Low quality streams damage guest confidence	In place
Headset sessions are synced	Hotel-side VR needs controlled device state	In place
PMS integration	Production hotels need booking system synchronization	Planned
Conversion analytics	Hotels need to measure preview impact on booking	Planned

FINAL PRODUCT DESCRIPTION

VirtuTour is a hospitality VR preview and virtual booking platform. It gives hotels a way to show guests the room before the booking is finalized, using 360 degree views, browser-based pixel streaming, and headset-ready immersive previews.

The platform connects hotel inventory, room media, UE5 production, streaming infrastructure, virtual booking queues, headset operations, and hotel-side monitoring into one workflow. The result is a direct bridge between visual trust and booking confirmation.

With VirtuTour, a hotel room is no longer represented only by a few static photos. A guest can explore the space, understand the view and layout, confirm the quality, and then book with confidence.